

Expression Linker

File path: Assets/UIAnimationToolkit/Editor/Animation/Anim_ExpressionLinker.cs

Menu: Tools > UI Animation Toolkit > Animation > Expression Linker

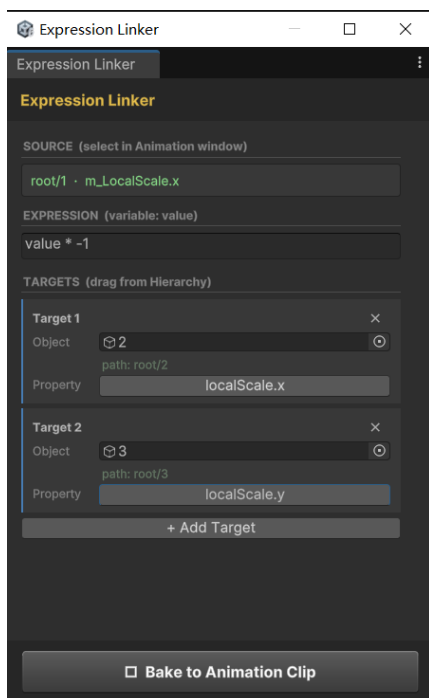
Overview

Transform one selected source property curve with a mathematical expression and bake the result to one or more target GameObjects and properties. The variable value represents a source key's value. Key values and tangents are transformed, making this useful for mirrored or proportionally linked animation.

How to use

1. Select a source property row in the Animation window. Check the SOURCE area for its binding path and property.
2. Enter a formula in EXPRESSION, such as `value * -1` or `value * 0.5 + 100`. A red field indicates an invalid expression.
3. Click + Add Target, assign a scene GameObject and choose a destination property. Add more targets if needed.
4. Click Bake to Animation Clip. Each source key is evaluated and written to the target curves; tangents are scaled using the expression derivative.

Interface and parameters



Supplied Expression Linker screenshot with two target properties.

Control / parameter	Description
SOURCE	Selected source binding path and property name.
EXPRESSION	Formula using value, +, -, *, / and parentheses.
Target Object	Drag a GameObject from the Hierarchy; its path is resolved relative to the clip root.
Target Property	Choose an exposed destination, including anchoredPosition.x/y, localScale.x/y, localEulerAngles.z, alpha and color channels.
+ Add Target	Append another destination entry.
Target remove button	Remove that target entry.
Bake to Animation Clip	Calculate and write target curves; Undo is recorded.

Expression examples

Expression / target	Result
value * -1	Mirror the values around zero.
value * 0.5	Halve the values.
value + 200	Add a constant offset of 200 units.
(value - 0.5) * 2	Center on 0.5 and double the result.
color.rgba target	Expand the target into four separate color-channel curves.

Notes

- Use the literal variable value. This is a small arithmetic expression parser, not an After Effects expression engine.
- Targets need the corresponding component/property. Inspect the destination before baking because its existing curve can be replaced.
- This creates baked editor data, not a live runtime constraint between objects.